$$\text{two zero entries: } \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{one zero entry: } \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \qquad \text{no zero entry: } \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

The following computations then show that every nontrivial ideal of R must contain one of the four matrices with only one nonzero entry, and hence must be all of R:

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

28. Gröbner Bases for Ideals

1. $-3x^3 + 7x^2y^2z - 5x^2yz^3 + 2xy^3z^5$

2. $-4x + 5y^3z^3 + 3y^2z^5 - 8z^7$

3. $2x^2yz^2 - 2xy^2z^2 - 7x + 3y + 10z^3$

4. $-8xy - 4xz + 3yz^3 + 2yz + 38$

5. $2z^5y^3x - 5z^3yx^2 + 7zy^2x^2 - 3x^3$

6. $-8z^7 + 3z^5y^2 + 5z^3y^3 - 4x$

7. $10z^3 - 2z^2y^2x + 2z^2yx^2 + 3y - 7x$

8. $3z^3y + 2zy - 4zx - 8yx + 38$

9. (See the answer in the text.)

10. $2xy^3z^5 - 5x^2yz^3 + 7x^2y^2z - 3x^3$

11. $3y^2z^5 - 8z^7 + 5y^3z^3 - 4x$

12. $2x^2yz^2 - 2xy^2z^2 + 10z^3 - 7x + 3y$

13. $3yz^3 - 8xy - 4xz + 2yz + 38$

14. We write the given ideal as $\langle x^2y + 4xy, xy^2 - 2x, xy - y^2\rangle$ so that the polynomials are listed in decreasing order; the maximum order term is x^2y. Multiplying the third by $-x$ and adding to the first (or dividing the first by the third), we write the ideal as $\langle xy^2 + 4xy, xy^2 - 2x, xy - y^2\rangle$ with maximum order term $xy^2 < x^2y$.

15. We write the given ideal as $\langle xy + y^3, x - y^4, y^3 + z\rangle$ so that the polynomials are listed in decreasing order; the maximum order term is xy. Multiplying the second by $-y$ and adding to the first (or dividing the first by the second), we write the ideal as $\langle y^5 + y^3, x - y^4, y^3 + z\rangle = \langle x - y^4, y^5 + y^3, y^3 + z\rangle$ with maximum order term $x < xy$.

16. We write the given ideal as $\langle x^3 + y^2z^3, x^2yz^3 + 4, xyz - 3z^3\rangle$ so that the polynomials are listed in decreasing order; the maximum order term is x^3. Because the leading terms of the second and third polynomials do not divide x^3, we cannot perform a single-step division algorithm reduction that gives a basis with a smaller maximum term order.

17. We write the given ideal as $\langle y^3z^2 - 2z, y^2z^3 + 3, y^2z^2 + 3\rangle$ so that the polynomials are listed in decreasing order; the maximum order term is y^3z^2. Multiplying the third by $-y$ and adding to the first (or dividing the first by the third), we write the ideal as $\langle -3y - 2z, y^2z^3 + 3, x^2z^2 + 3\rangle = \langle y^2z^3 + 3, y^2z^2 + 3, -3y - 2z\rangle$ with maximum order term $y^2z^3 < y^3z^2$.

18. Starting with the given basis and adding (-2)(1st) to the 2nd, and adding (-1)(1st) to the 3rd yields

$$\langle w + x - y + 4z - 3, -x + 3y - 10z + 10, 2x - 2y - 3z - 2\rangle.$$

Adding (2)(2nd) to the 3rd yields

$$\langle w + x - y + 4z - 3, -x + 3y - 10z + 10, 4y - 23z + 18\rangle$$

Thus $\{w + x - y + 4z - 3, -x + 3y - 10z + 10, 4y - 23z + 18\}$ is a Gröbner basis.

19. Starting with the given basis and adding (-2)(1st) to the 2nd and adding (-1)(1st) to the 3rd yields

$$\langle w - 4x + 3y - z + 2, 6x - 5y + 1, -6x + 5y - 3\rangle.$$

Adding the 2nd to the 3rd yields

$$\langle w - 4x + 3y - z + 2, 6x - 5y + 1, -2\rangle.$$

Because the ideal contains a unit, -2, we know that the ideal is equal to $\mathbb{R}[w, x, y, z] = \langle -2\rangle = \langle 1\rangle$. Thus any set $\{a\}$, where $a \neq 0$ and $a \in \mathbb{R}$, is a Gröbner basis. (The corresponding algebraic variety is $\varnothing$.)

20. Because every ideal in $\mathbb{R}[x]$ is principal, a Gröbner basis will consist of a polynomial of minimum degree that has the form $f(x)$(1st) + $g(x)$(2nd). Now when we add $(-x)$(2nd) to the 1st, we obtain $\langle x^2 - 4, x^3 + x^2 - 4x - 4\rangle$. Adding $(-x)$(1st) to the 2nd, we obtain $\langle x^2 - 4, x^2 - 4\rangle = \langle x^2 - 4\rangle$. Thus $\{x^2 - 4\}$ is a Gröbner basis.

21. Because every ideal in $\mathbb{R}[x]$ is principal, a Gröbner basis will consist of a polynomial of minimum degree that has the form $f(x)$(1st) + $g(x)$(2nd) + $h(x)$(3rd). Adding $(-x)$(2nd) to the 1st and adding (-1)(2nd) to the 3rd, we obtain

$$\langle -3x^3 + 9x^2 - 6x, x^3 - x^2 - 4x + 4, x^2 + x - 2\rangle.$$

Adding $(3x)$(3rd) to the 1st and adding $(-x)$(3rd) to the 2nd yields

$$\langle 12x^2 - 12x, -2x^2 - 2x + 4, x^2 + x - 2\rangle.$$

Adding (-12)(3rd) to the 1st and adding (2)(3rd) to the 2nd yields

$$\langle -24x + 24, 0, x^2 + x - 2\rangle = \langle x - 1, (x - 1)(x + 2)\rangle = \langle x - 1\rangle.$$

Thus $\{x - 1\}$ is a Gröbner basis.

22. Because every ideal in $\mathbb{R}[x]$ is principal, a Gröbner basis will consist of a polynomial of minimum degree that has the form $f(x)$(1st) + $g(x)$(2nd). Now when we add $(-x^2)$(2nd) to the 1st, we obtain $\langle x^4 - x^3 + 2x^2 + 2x - 5, x^3 - x^2 + x - 1\rangle$. Adding $(-x)$(2nd) to the 1st yields $\langle x^2 + 3x - 5, x^3 - x^2 + x - 1\rangle$. Adding $(-x)$(1st) to the 2nd yields $\langle x^2 + 3x - 5, -4x^2 + 6x - 1\rangle$. Adding (4)(1st) to the 2nd yields $\langle x^2 + 3x - 5, 18x - 21\rangle$. Adding $(-x/18)$(2nd) to the 1st, we obtain $\langle \frac{25}{6}x - 5, 18x - 21\rangle = \langle \frac{5}{6}x - 1, 6x - 7\rangle$. Adding (-36/5)(1st) to the 2nd yields $\langle \frac{5}{6}x - 1, \frac{1}{5}\rangle = \langle 1\rangle$. Thus $\{1\}$ is a Gröbner basis. (The corresponding algebraic variety is $\varnothing$.)

23. Adding $(-x)$(2nd) to the 1st, we get $\langle -2xy + 8x - 2, xy + 2y - 9\rangle$. Adding (2)(2nd) to the 1st yields $\langle 8x + 4y - 20, xy + 2y - 9\rangle = \langle 2x + y - 5, xy + 2y - 9\rangle$. Adding $(-y/2)$(1st) to the 2nd yields $\langle 2x + y - 5, -\frac{1}{2}y^2 + \frac{9}{2}y - 9\rangle = \langle 2x + y - 5, y^2 - 9y + 18\rangle = \langle g_1, g_2\rangle$.

We now proceed to test for a Gröbner basis according to Theorem 28.12. Maximum term degree can't be reduced by the division algorithm. Form $S(g_1, g_2) = (y^2)(1st) - (2x)(2nd) = 18xy - 36x + y^3 - 5y^2$. This is of greater term order than either g_2 or g_1. We see if this can be reduced to zero using g_1 and g_2, that is, repeatedly using the division algorithm on remainders with just g_1 or g_2 as divisors. We have

$$\begin{aligned} S(g_1, g_2) - 9y(g_1) &= 18xy - 36x + y^3 - 5y^2 - 9y(2x + y - 5) \\ &= -36x + y^3 - 14y^2 + 45y. \end{aligned}$$

We add $18g_1 = 36x + 18y - 90$ to this and obtain

$$y^3 - 14y^2 + 63y - 90.$$

We add $(-y)g_2 = -y^3 + 9y^2 - 18y$ to this and obtain

$$-5y^2 + 45y - 90.$$

Finally, adding $5g_2 = 5y^2 - 45y + 90$ to this we obtain 0. Thus by Theorem 18.12, we see that $\{g_1, g_2\} = \{2x + y - 5, y^2 - 9y + 18\}$ is a Gröbner basis.

Because $y^2 - 9y + 18 = (y - 6)(y - 3)$, any point on the corresponding variety has y-coordinate 3 or 6. Requiring that the point be a zero of $2x + y - 5$, we find that the variety is $\{(1, 3), (-\frac{1}{2}, 6\}$.

24. Let $N = \langle x^2y + x, xy^2 - y\rangle = \langle g_1, g_2\rangle$. Maximum term degree in the basis $\{g_1, g_2\}$ cannot be reduced further by using the division algorithm. We compute $S(g_1, g_2) = yg_1 - xg_2 = y(x^2y + x) - x(xy^2 - y) = 2xy$, and proceed to reduce the basis $\{x^2y + x, xy^2 - y, xy\}$ using the division algorithm. Adding $(-x)$(3rd) to the 1st and adding $(-y)$(3rd) to the 2nd, we obtain $N = \langle xy, x, y\rangle = \langle x, y\rangle$. Thus $\{x, y\}$ is a Gröbner basis and the corresponding algebraic variety is the origin, $\{(0, 0)\}$.

25. Let $N = \langle x^2y + x + 1, xy^2 + y - 1\rangle = \langle g_1, g_2\rangle$. Maximum term degree in the basis $\{g_1, g_2\}$ cannot be reduced further by using the division algorithm. We therefore compute $S(g_1, g_2) = y(x^2y + x + 1) - x(xy^2 + y - 1) = x + y$, and proceed to reduce the basis $\{x^2y + x + 1, xy^2 + y - 1, x + y\}$ using the division algorithm. Adding (-xy)(3rd) to the 1st, we obtain $N = \langle -xy^2 + x + 1, xy^2 + y - 1, x + y\rangle$. Adding the 2nd to the 1st, we have $N = \langle xy^2 + y - 1, x + y\rangle$. Adding $(-y^2)$(2nd) to the 1st, we obtain $N = \langle x + y, -y^3 + y - 1\rangle = \langle h_1, h_2\rangle$ where maximum term degree cannot be reduced further using the division algorithm.

While we can determine the algebraic variety easily now, for illustration, we verify that we do have a Gröbner basis using Theorem 28.12. Now $S(h_1, h_2) = y^3h_1 + xh_2 = xy - x + y^4 = x(y - 1) + y^4$, and we test if it can be reduced to zero using the division algorithm with h_1 and h_2 as divisors. We obtain $(1 - y)h_1 + x(y - 1) + y^4 = (1 - y)(x + y) + x(y - 1) + y^4 = y^4 - y^2 + y$. Adding yh_2 to this yields 0, so by Theorem 28.12, we see that $\{h_1, h_2\} = \{x + y, -y^3 + y - 1\}$ is a Gröbner basis for N.

Using our calculator, we find that $y^3 - y + 1$ has one real zero which is approximately -1.3247, so $V(N) = \{a, -a)\}$ for $a \approx 1.3247$.

26. Let $N = \langle x^2y + xy^2, xy - x\rangle$. Adding $(-x)$(2nd) to the 1st, we discover that $N = \langle x^2 + xy^2, xy - x\rangle = \langle g_1, g_2\rangle$ where maximum term degree cannot be reduced further using the division algorithm. We compute $S(g_1, g_2) = y(x^2 + xy^2) - x(xy - x) = x^2 + xy^3$, and test if it can be reduced to zero using

the division algorithm repeatedly with g_1 and g_2 as divisors. We find that $(x^2+xy^3)+(-1)g_1 = (x^2+xy^3)-(x^2+xy^2) = xy^3-xy^2$. Adding $(-y^2)g_2 = (-y^2)(xy-x)$ to xy^3-xy^2 yields 0, so $\{x^2+xy^2, xy-x\}$ is a Gröbner basis for N by Theorem 28.12. We can obtain a slightly simpler Gröbner basis by adding $(-y)$(2nd) to the 1st, which yields $\{x^2+xy, xy-x\}$. Then adding (-1)(2nd) to the 1st gives us $\{x^2+x, xy-x\}$ as Gröbner basis. Note that the initial power products x^2 and xy of the basis polynomials remain the same.

Now x^2+x has 0 and -1 as zeros. We see that $(0,a)$ is a zero of $xy-x$ for all $a \in \mathbb{R}$, but $(-1,b)$ is a zero of $xy-x$ only if $b=1$. Thus $V(N) = \{(-1,1), (0,a) \mid a \in \mathbb{R}\}$.

27. T F T T T T T T F F (The answer T to Part(**h**) assumes that the student has had a course in linear algebra where matrix reduction was used to solve linear systems.)

28. Let $P_i = xy$ and $P_j = x^2$ in lex with $y < x$. Then $xy < x^2$ but xy does not divide x^2.

29. *Additive closure:* With $c_i, d_i \in R$ for $i = 1, 2, \cdots, r$, we have

$$(c_1f_1 + c_2f_2 + \cdots + c_rf_r) + (d_1f_1 + d_2f_2 + \cdots, +d_rf_r) = (c_1+d_1)f_1 + (c_2+d_2)f_2 + \cdots + (c_rf_r + d_rf_r)$$

and $(c_i + d_i)$ is in R for $i = 1, 2, \cdots, r$.

Additive identity: Set all $c_i = 0$

Additive inverses: Replace all c_i by $-c_i$.

Multiplicative property: For $a \in R$, $a(c_1f_1 + c_2f_2 + \cdots + c_rf_r) = (ac_1)f_1 + (ac_2)f_2 + \cdots + (ac_r)f_r$ and $ac_i \in R$ for $i = 1, 2, \cdots, r$.

30. Let $h(\mathbf{x})$ be a common divisor of $f(\mathbf{x})$ and $g(\mathbf{x})$, so that $f(\mathbf{x}) = q_1(\mathbf{x})h(\mathbf{x})$ and $g(\mathbf{x}) = q_2(\mathbf{x})h(\mathbf{x})$ for $q_1(\mathbf{x})$, $q_2(\mathbf{x})$ in $F[\mathbf{x}]$. Then

$$\begin{aligned} r(\mathbf{x}) = f(\mathbf{x}) - g(\mathbf{x})q(\mathbf{x}) &= q_1(\mathbf{x})h(\mathbf{x}) - q_2(\mathbf{x})h(\mathbf{x})q(\mathbf{x}) \\ &= [q_1(\mathbf{x}) - q_2(\mathbf{x})q(\mathbf{x})]h(\mathbf{x}), \end{aligned}$$

so $h(\mathbf{x})$ divides $r(\mathbf{x})$ as well as $g(\mathbf{x})$. Thus divisors of both $f(\mathbf{x})$ and $g(\mathbf{x})$ also divide both $g(\mathbf{x})$ and $r(\mathbf{x})$.

Going the other way, suppose $k(\mathbf{x})$ divides both $g(\mathbf{x})$ and $r(\mathbf{x})$, so that $g(\mathbf{x}) = q_3(\mathbf{x})k(\mathbf{x})$ and $r(\mathbf{x}) = q_4(\mathbf{x})k(\mathbf{x})$ for $q_3(\mathbf{x})$, $q_4(\mathbf{x}) \in F[\mathbf{x}]$. Then

$$\begin{aligned} f(\mathbf{x}) = g(\mathbf{x})q(\mathbf{x}) + r(\mathbf{x}) &= q_3(\mathbf{x})k(\mathbf{x})q(\mathbf{x}) + q_4(\mathbf{x})k(\mathbf{x}) \\ &= [q_3(\mathbf{x})q(\mathbf{x}) + q_4(\mathbf{x})]k(\mathbf{x}), \end{aligned}$$

so $k(\mathbf{x})$ divides $f(\mathbf{x})$ as well as $g(\mathbf{x})$. Thus divisors of both $g(\mathbf{x})$ and $r(\mathbf{x})$ also divide both $g(\mathbf{x})$ and $f(\mathbf{x})$.

Thus the set of divisors of $f(\mathbf{x})$ and $g(\mathbf{x})$ is the same as the set of divisors of $g(\mathbf{x})$ and $r(\mathbf{x})$.

31. Let $N = \langle xy, y^2-y\rangle = \langle g_1, g_2\rangle$. By Theorem 28.12, we need to show that $S(g_1,g_2) = y(xy) - x(y^2-y) = xy$ can be reduced to 0 using the division algorithm with just xy and y^2-y as divisors. Adding $(-1)g_1$ to xy, we immediately obtain 0, and we are done.

32. *Additive closure:* Let $f(\mathbf{x})$, $g(\mathbf{x}) \in I(S)$, so that $f(\mathbf{s}) = 0$ and $g(\mathbf{s}) = 0$ for all $\mathbf{s} \in S$. Applying the evaluation homomorphism $\phi_{\mathbf{s}}$, we get $\phi_{\mathbf{s}}(f(\mathbf{x}) + g(\mathbf{x})) = \phi_{\mathbf{s}}(f(\mathbf{x})) + \phi_{\mathbf{s}}(g(\mathbf{x})) = f(\mathbf{s}) + g(\mathbf{s}) = 0 + 0 = 0$ for all $\mathbf{s} \in S$, so $(f+g) \in I(S)$.